



FIRST TERM EXAMINATION 2026

CLASS M-II

SUBJECT BIOLOGY

Section -A (Marks 11)

Time Allowed: 15 Minutes

Section-A is compulsory. All parts of this section are to be answered on this page and handed over to the invigilator. Deleting/overwriting is not allowed. Do not use lead pencil.

Do not use lead pencil.

Student Name: cheemab Ahif

Section: MII-H

Roll Number						
7	4	6	8	3		
0	0	0	0	0	0	0
1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Section: Math - P

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	E
1. The sum of two numbers is 10. One number is 6. What is the other number?	☐	☐	☐	☐	☐
2. A rectangle has a length of 8 cm and a width of 5 cm. What is its perimeter?	☐	☐	☐	☐	☐
3. A triangle has angles of 30°, 60° and 90°. What is the longest side?	☐	☐	☐	☐	☐
4. A car travels at a constant speed of 60 km/h. How far will it travel in 2 hours?	☐	☐	☐	☐	☐
5. A box contains 10 red balls and 5 blue balls. What is the probability of drawing a red ball?	☐	☐	☐	☐	☐
6. A circle has a radius of 7 cm. What is its area?	☐	☐	☐	☐	☐
7. A right-angled triangle has sides of 3 cm, 4 cm and 5 cm. What is the hypotenuse?	☐	☐	☐	☐	☐
8. A square has a side length of 6 cm. What is its diagonal?	☐	☐	☐	☐	☐
9. A cylinder has a radius of 3 cm and a height of 10 cm. What is its volume?	☐	☐	☐	☐	☐
10. A cone has a radius of 4 cm and a height of 6 cm. What is its surface area?	☐	☐	☐	☐	☐

A diagram of a cell. A large circle represents the cell membrane. Inside, a smaller circle represents the nucleus. Inside the nucleus is a small, dense, dark circle representing the nucleolus. Arrows point from labels 'Q' and 'R' to the nucleolus, and an arrow points from label 'P' to the nucleus.

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Signature Of Invigilator

Copy No:

Name: Cheemab Arif

Roll No. 24683

Class: M-2

Section: 11

Subject: Biology

Date: 27-Aug-2026 No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained	6.5	22	6.5	6.5									41	50

Start writing here

Section :- B

Short Answer :-

Q.2(i)

A.2(i)

① Paleontology :- Paleontology is the study of fossil fuels. The importance of this branch is to know about fossil fuels, their types and structures.

② Molecular biology :- Molecular biology is the study of molecular levels. The main point of this branch is to study about molecules, bio-molecules like proteins, lipids, carbohydrates and nucleic acids, their ~~store~~ structural and functional features.

③ Immunology :- Immunity is when the body itself is protecting against

foreign substances like microbes ~~to~~ and the study of immunity is immunology. The main significance of immunology is to protect yourself from diseases.

Q:2(ii)

A:2(ii)

Structural conformation :-

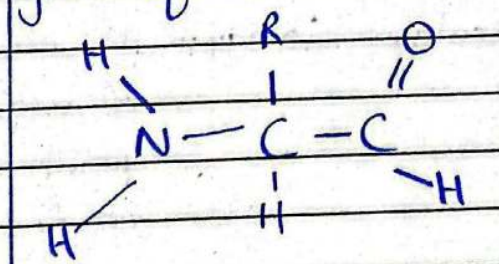
Structure of proteins are classified into primary, secondary, tertiary and or in some cases quaternary. A linear polypeptide chains folds ~~into~~ in a number, specific type or sequence is called primary structures.

After synthesis, these proteins does not remain in primary structures. ~~they~~ they are folded into higher level bonding like secondary, quaternary or tertiary.

Amino acids:-

Amino acids are the building blocks of ~~proteins~~ or monomers. They are commonly found in proteins. There are 20-25 different types of amino acids.

Each amino acids have a central carbon atom called α (alpha) carbon. and there are different groups



attached with alpha carbon atom including; Hydrogen group (-H), an amino group (-NH₂), a carboxyl group (-COOH) and an ~~alkyl~~ alkyl group (-R) known as functional group.

Q: 2(iii)(or)

A: 2(iii)(or)

Collaboration ^{helps} in scientists :- (02)

Collaboration really help scientists in making researches and discovery. In Covid era, Scientists helped each other, some have same knowledge whereas other have different knowledge. This collaboration is known as interdisciplinary collaborative research.

Q: 2(iv)(or)

A: 2(iv)(or)

* Types of RNA:-

① mRNA:- mRNA sends message to ribosome and these ~~at~~ ribosomes transfer amino acids in the form of ~~to~~ protein. Synthesis is known mRNA. This is used in translation.

② tRNA :- tRNA {transfer ribonucleic acid} transfer amino acids to ribosomes.

③ rRNA :- rRNA {ribosomal ribonucleic acid} is the component of RNA. ~~and are used~~ ^{ribosomes}

Q: 2(v)(or)

A: 2(v)(or)

main types of ~~at~~ plastids:-

Chloroplast:- are ~~is~~ used for making food for plant.

Chromoplast:- are used to get attention from birds for seed dispersal.

02

Leucoplast:- are used as food storage.

Q:2(vi)

A:2(vi)

Respiratory and circulatory system:-

Organs are working together i.e. respiratory and circulatory system { cardiovascular system } in which circulatory system pumps blood to all body and respiratory system absorb oxygen and eliminate carbon dioxide.

g.s

Q:2(vii)

A:2(vii)

1.5

Meiosis serve as the basis of asexual reproduction because, in asexual reproduction plants, fungi and bacteria only make 2 diploid daughter cells and they have same chromosomes as their parent cells.

Q:2(viii)(or)

A:2(viii)(or)

Domain Bacteria

Domain Eukarya

Prokaryotic or eukaryotic cells.

① Domain Bacteria are prokaryotic cells.

① Domain Eukarya are eukaryotic cells.

cell wall

presence

② Cell wall of Bacteria contains peptidoglycan.

Not all eukarya have cell wall, cell wall of eukarya have cellulose

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Signature Of Invigilator

[Signature]

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Roll No. 24683

Class: M-2 Section: N Subject: Biology

Date: 27-8-2020 No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained														

Start writing here

rRNA uniqueness.

- ③ Their rRNA {ribosomal RNA} is only unique to Bacteria to eukarya.

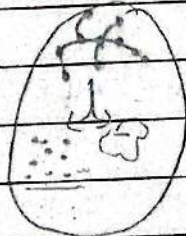
03

Antibiotic effectiveness.

- ④ Bacteria is ~~not~~ resistant to antibacterial ^{traditional} antibiotics. They are sensitive to some antibiotic that affect bacteria.
- ④ Eukarya is sensitive to ~~antibacterial~~ ^{antibacterial} traditional antibiotics that affect they are resistant to some antibiotics that affect eukarya.

Q.8(in)

A:2(in)



03



* Biochemical reaction that ~~molecules~~ are larger and complex molecules are broken down into smaller and simpler parts

Biochemical reactions that are ~~complex~~ molecules build from small and simpler parts are called anabolism or

are called catabolism or catabolic reaction.
catabolic reactions

for Example:- In digestion, is a process which is converting complex and larger foods into smaller molecules.

for Example:- In photosynthesis, is a process which is made by CO_2 from glucose converted from CO_2 from air and H_2O from soil.

Q:3(i)

A:3(i)

Enzyme Activity:-

Temperature:- Each enzyme increase with increase of temperature but up to a specific limit.

Each enzyme works best at a specific temperature, which is called optimum temperature. Different enzyme have different optimum temperature. for Example:-

Optimum temperature for Human enzyme is 36°C to 38°C . Any increase or decrease effect of temperature effect the rate of enzymatic reaction.

- ① At a very high temperature:- Atoms of an enzyme molecule starts vibrating vigorously and its active site is damaged and it is called denatured. At a very low temperature:- enzyme molecules remain inactive due to fall in activation energy. They will start their catalyzing activity once their normal temperature are restored.

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Signature Of Investigator

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Name: Shamsh Ali

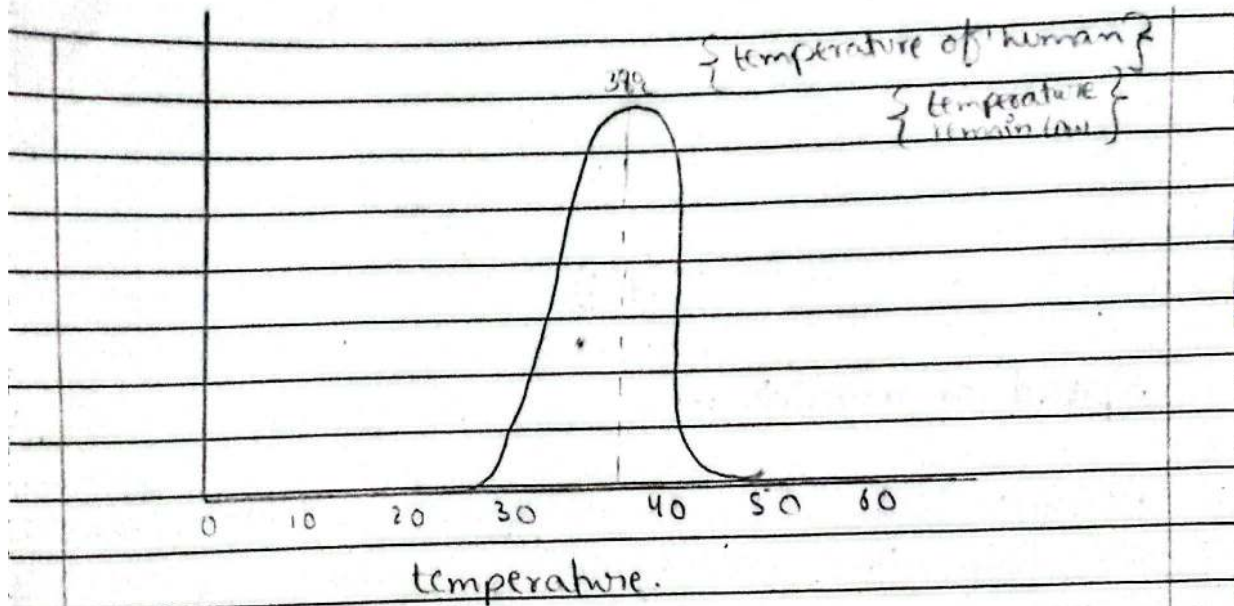
Roll No. 24683

Class: M-2 Section: 11 Subject: Biology

Date: 27-August-26 No Of Continuation Sheets Attached:

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
Marks Obtained														

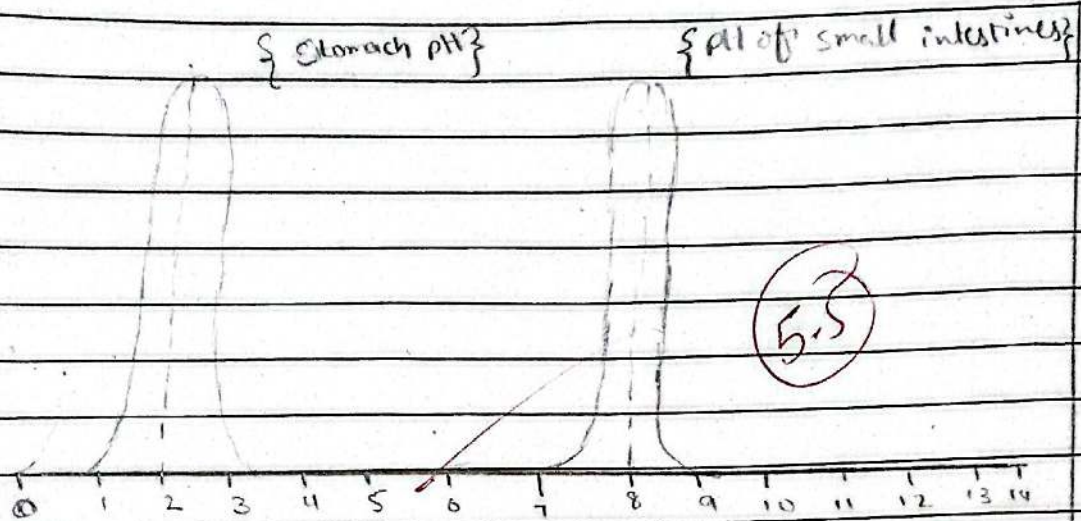
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pH:- Each enzyme have affecting ^{medium} pH. Each enzyme work best at specific pH, which is called optimum pH. Slight change in affecting pH ~~medium~~, below or above the optimum pH can damaged active site. Extreme changes in pH can damaged the enzyme molecule. for Example:-

Pepsin and trypsin are both protein digesting enzyme. Pepsin enzyme work best ~~at~~ in acidic medium in stomach at 2.00 pH, whereas trypsin enzyme works best in alkaline medium in small intestine

at 8.00 pH.



Q.4(i)
A.4(i)

Spread of malaria :-

Malaria is a fatal disease from ancient times now we know that the cause of malaria is Plasmodium but scientists still do not know ~~what agent is~~ how Plasmodium is in malarial patient's blood. Scientists have to find that agent.

Observation :- A.F.A. King has 2 observations in 19th century. His observations were :-

- ① Drinking marshes water doesn't cause malaria.
- ② Those who slept indoors are not likely to get malaria than those who slept outdoors.

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Signature Of Invigilator

[Signature]

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Name: Seemab Afif

Roll No. 24683

Class: M-2

Section tl

Subject Biology

Date: 27-Aug-2020

No Of Continuation Sheets Attached

Question No.	1	2	3	4	5	6	7	8	9	10	11	12	Total	Out of
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Start writing here

③ Those who slept under nets are likely not to get than ~~who~~ those who didn't slept under nets are likely to get malaria.

④ Sleeping near smokes does not cause ~~not~~ malaria.

Hypothesis:- Hypothesis is 'mosquitoes is transmitting the plasmodium from malaria person'.

Deduction ::

Deduction I ::

If mosquitoes is transmitting plasmodium ~~from malaria~~ to patients. then all patients have plasmodium in their blood.

Deduction II ::

If mosquitoes is transmitting plasmodium from one area to another then it must be ~~transmity~~ transmitting from malarial patients.

Experimentation:- A German chemist, A.F.A. King performed this experiment in India while showing that.

When he ~~was~~ captured one malarial sparrow and one healthy one, he made Culex {female mosquito} bite that malarial sparrow but after sometime he made it bite to healthy bird and he killed that mosquitoes. After sometimes, the healthy sparrows ^{are} becoming malarial sparrows.

He did same with Anopheles mosquitoes and the results were same.

In result:- He made it clear that ~~Plasmodium~~ mosquitoes are transmitting plasmodium from malarial ill patients and ~~and~~ animals and the hypothesis ~~were~~ true

5.5